1 May 2003

TO: Pre–Installation Tab 088

SUBJECT: **DIRECTION 2294003, SIGNA Twin*Speed* PRE–INSTALLATION, REV 3**

Enclosed is *Direction 22294003, Signa TwinSpeed Pre–Installation,* Rev 3 revision. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2294003 REVISION 3 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
Throughout manual	• Added BrainWave Option For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY, information includes new revised heat output values.
1 SYSTEM CONFIGURATION	• Table 1–11 Note 2 changed M3043TA to M3043TC • Deleted Remote Keyboard Option for use with Operator Workspace PC LCD catalogs.
2 ROOM LAYOUTS	• Illustration 2–33 corrected Magnet Monitor vertical box and mounting flange dimensions.
4 SITE ENVIRONMENT	• Table 4–2 **bold** Note 2 to emphasize air cooling load average of maximum values. • Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC Table 4–7 Note 6 added Ethylene Glycol or Propylene Glycol wording. • Section 4–4–5 Water Cooled TGWC Requirements Table 4–8 updated Pressure Drop values, added note clarifying value for water and factor for 50/50, and Illustration 4–4 Water Flowrate/ Temperature requirement. • Section 4–7–3 Design Guidelines for acoustics added Structural Vibration Control information. • Section 4–8 Room Ventilation pressure equaliztion waveguide requirement added IEC regulation reference. •
5 POWER REQUIREMENTS	• Section 5–3–1 MDP and Illustration 5–1 clarified Emergency Off buttons provided with GE MDP option. • Illustration 5–1 added SPLICE on Run 296 BLK wire connection. • Illustration 5–3 external conduit metal requirement added 2002 NEC reference, also updated RF Filter metaliic conduit requirement and filter location for 2002 NEC requirement.
6 INTERCONNECT DATA	• Ilustration 6–1 added Equipment Room and Control Area delineating lines (requested by Installation Specialist). • Illustration 6–1, Tables 6–2 & 6–3 corrected cable Group 81 to connect between E02 and PP1 • Group 25 added missing Run 726 • Table 6–3 (multiple pages) added M3044TC to notes for increased usable length. • RF Transmit interconnects changed for new material & standard length cut at site. Added Group 72 from MR1 to MG2 for with RF Transmits (removed from Groups 25 & 45).
9 PREINSTALLATION	• Changed hospital to facility in several items.
10 TOOLS & TEST EQUIPMENT	• Deleted RF survey apparatus no longer available from GE Medical Systems.

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2294003 REV 1

Refer to Table 2 for instructions to properly update your Direction 2294003 Rev 2 with the enclosed Revision 3 pages.

TABLE 2
DIRECTION 2294003 REVISION 2 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
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	A/B	A/B
	i to ix	i to ix
1 SYSTEM CONFIGURATION	1–9/10 1–13/14	1–9/10 1–13/14
2 ROOM LAYOUT	2–1 to 2–71 (entire section)	2–1 to 2–75 (entire section)
4 SITE ENVIRONMENT	4–1 to 4–40 (entire section)	4–1 to 4–43 (entire section)
5 POWER REQUIREMENTS	5–1 to 5–18 (entire section)	5–1 to 5–18 (entire section)
6 INTERCONNECT DATA	6–1 to 6–43 (entire section)	6–1 to 6–45 (entire section)
8 SHIPPING & DELIVERY DATA	8–5/6	8–5/6
9 PREINSTALLATION	9–3/4 9–7/8	9–3/4 9–7/8
10 TOOLS & TEST EQUIPMENT	10–1 to 10–10 (entire section)	10–1 to 10–10 (entire section)

Debra C. Balis
MR SERVICE ENGINEERING

GE Medical Systems

Technical Publications

Direction 2294003
Revision 3

Signa® 1.5T Twin*Speed*
Pre–Installation

Architectural/Installation Planning Use Only

Operating Documentation

GE Medical Systems

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WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Prelim A	Mar. 21, 2001	Initial Development Preliminary version. Information subject to change without notice
Rev 0	June 28, 2001	Initial Product Release
Rev 1	May 21, 2002	Added Signa TwinSpeed with EXCITE Technology system information throughout the document. Other major changes listed by section include the following **Sec 1:** Update power product offerings to E catalogs. **Sec 2:** Update TSCC & TGWC for revised E&W information; Added NEC 2002 Article references for MDP information; Added Network connection and telephone lines option to system connectivity requirements; Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogens; Updated TAC Cabinet dimensions drawing. **Sec 4:** Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications; Updated Water Cooled TSCC water cooling requirements; Clarified Magnet Room exhaust fan intake location requirement. **Sec 5:** Revised Voltage Transient specification; Updated Facility Ground Wire measurement techniques; Added NEC 2002 Article references. **Sec 6:** Updated miscellaneous system interconnects. **Sec 7:** Added illustration of Magnet Minimum Ceiling height area requirement; Clarified RF Shield Room vendor responsibility to supply torque specifications for all anchors, studs, and fastening hardware to meet the clamping force specified. **Sec 8:** Update magnet moving dimensions height. **Sec 9:** Updated references to Network connection and telephone lines option to system connectivity; Updated Cryogenic Vent connected to magnet prior to magnet ramp–up. **Sec 10:** Added new & deleted obsolete test equipment & part numbers. **Appendix C:** Added Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*" document.
Rev 2	Nov. 19, 2002	Direction 2128126 updated to Rev 1. Changed SRF Cabinet to 1.5T SRFD II Cabinet information throughout the manual. **Sec 1:** Added M1090ZP & M1090ZN OW ONLY Long Cables; System Cooling & MDP selections Water Cooled TGWC added MDP turnkey solution choice to flowchart illustration; Added UPS for Magnet Monitor to Mobile magnet catalogs; For TSCC catalogs table added note for MDP required; Facility Options added full system UPS and Bypass panel options catalogs. **Sec 2:** Added Multiple MR System Sites shared Equipment Room requirements; Cradle extensions information added for TRM Coil replacement; Mounting requirements clarified for MDP top breaker highest position; Added UPS for Magnet Monitor to Mobile magnet catalogs; Revised System Monitoring & Support Connectivity Requirements for Magnet Monitor; Updated ACGD/PDU Cabinet weight & center of gravity values. **Sec 3:** Corrected reference typographical error; Magnetic Isogauss line plots added 40mT & 200mT required by IEC 60601–2–33 6.8.3 bb. **Sec 4:** Revised hearing protection requirement in the Magnet Room; Clarified water cooling required 365 days per year for Water Cooled TSCC & Water Cooled TGWC; Added metric value for recommend Magnet Room ventilation; Exhaust Fan illustrations added Room Ventilation Requirements/Recommendations reference; Added roof hatch delivery situation to Cryogenic Vent inspection CAUTION; Typical Cryogenic Vent Detail added height requirement for RF Shielded Room Contractor supplied vent pipe/waveguide in Magnet Room. **Sec 5:** Protective Disconnect Set–Up clarified power recommendation for Shield Cooler & TSCC; Added PDU feeder wiring requirement with full system UPS option; Updated Common Ground Stud location dimension in Magnet Room Grounding Requirements illustration. **Sec 6:** Added notes to identified RF Transmit & RF Receive cables; Updated miscellaneous system interconnects; Cable Groups 8 & 12 revised for Gradient Cable length reduction. **Sec 7:** References typo error corrected; Added use of Megger Insulation Tester to Electrical Isolation Measurement Method for Anchor Hardware Requirements for MR Equipment Inside RF Shielded Room. **Sec 8:** Added Transportation Environmental Conditions. **Sec 10:** Added Resistance Meter to table of Rigger/Customer Supplied Equipment. **Appendix D:** Added REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS.
Rev 3	May 1, 2003	Added BrainWave Option For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY, information includes new revised heat output values. **Sec 1:** Table 1–11 Note 2 changed M3043TA to M3043TC; Deleted Remote Keyboard Option for use with Operator Workspace PC LCD catalogs. **Sec 2:** Illustration 2–33 corrected Magnet Monitor vertical box and mounting flange dimensions. **Sec 4:** Table 4–2 **bold** Note 2 to emphasize air cooling load average for maximum value; Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC Table 4–7 Note 6 added Ethylene Glycol or Propylene Glycol wording. Section 4–4–5 Water Cooled TGWC Requirements Table 4–8 updated Pressure Drop values, added note clarifying value for water and factor for 50/50, and Illustration 4–4 Water Flowrate/Temperature requirement. Section 4–7–3 Design Guidelines for acoustics added Structural Vibration Control information. Section 4–8 Room Ventilation pressure equaliztion waveguide requirement added IEC regulation reference. Section 4–14–4 Transient Vibration specification added clarifications for measurements. **Sec 5:** Section 5–3–1 MDP and Illustration 5–1 clarified Emergency Off buttons provided with GE MDP option; Illustration 5–1 added SPLICE on Run 296 BLK wire connection & clarified Emergency OFF. Illustration 5–3 external conduit metal requirement added 2002 NEC reference, also updated RF Filter metaliic conduit requirement and filter location for 2002 NEC requirement. **Sec 6:** Ilustration 6–1 added Equipment Room and Control Area delinating lines (requested by Installation Specialist); Illustration 6–1, Tables 6–2 & 6–3 corrected cable Group 81 to connect between E02 and PP1Group 25 added missing Run 726; Table 6–3 (multiple pages) added M3044TC to notes for incresed usable length; RF Transmit interconnects changed for new material & standard length cut at site. Added Group 72 from MR1 to MG2 for with RF Transmits (removed from Groups 25 & 45). **Sec 9:** Changed hospital to facility in several items. **Sec 10:** Deleted RF survey apparatus no longer available from GE Medical Systems.

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GLOSSARY

ACGD – Abbreviation for Advanced Control Gradient Drivers.

CRYOGEN – A substance for producing low temperatures. Liquid helium is the cryogen used to cool the magnet to approximately 4 Kelvin (–269°C or –452°F).

CRYOSTAT – An apparatus maintaining a very low constant temperature. The cryostat consists of one concentric, cylindrical container housed in an outer vacuum tight vessel. The magnet and shim coils are mounted in the inner container. The container is filled with liquid helium. The shields surrounding the inner container are kept cold by a refrigeration device.

DEWAR – A container with an evacuated space between two highly reflective walls used to keep low temperature substances at near–constant temperatures. Liquid helium is usually stored and shipped in dewars.

EXCLUSION ZONE – Area where the magnetic flux density is greater than five gauss. Personnel with cardiac pacemakers, neurostimulators and other biostimulation devices must NOT enter this zone. Signs are posted outside the five gauss line alerting personnel of this requirement. Since the magnetic field is three–dimensional, signs are also posted on floors above and below the Magnet Room in which the five gauss line exists.

FERROUS MATERIAL – Any substance containing iron which is strongly attracted by a magnetic field.

GAUSS (G) – A unit of magnetic flux density. The earth's magnetic field strength is approximately one half gauss to one gauss depending on location. The internationally accepted unit is the tesla (1 Tesla = 10,000G and 1 milli Tesla = 10G).

GRADIENT – The amount and direction of the rate of change in space of the magnetic field strength. In the magnetic resonance system, gradient amplifiers and coils are used to vary the magnetic field strength in the x, y, and z planes.

HOMOGENEITY – Uniformity. The homogeneity of the static magnetic field is an important quality of the magnet.

ISOCENTER – Center of the imaging volume ideally located at the magnet center.

ISOGAUSS LINE – An imaginary line or a line on a field plot connecting identical magnetic field strength points.

MAGNETIC FIELD (H) – A condition in a region of space established by the presenece of a magnet and characterized by the presence of a detectable magnetic force at every point in the region. A magnetic field exists in the space around a magnet (or current carrying conductor) and can produce a magnetizing force on a body within it.

MAGNETIC RESONANCE (MR) – The absorption or emission of electromagnetic energy by nuclei in a static magnetic field, after excitation by a suitable radio frequency field.

MAGNETIC SHIELDING – Using material (e.g. steel) to redistribute a magnetic field , usually to reduce fringe fields.

QUENCH – Condition when a superconducting magnet becomes resistive thus rapidly boiling off liquid helium. The magnetic field reduces rapidly after a quench.

GLOSSARY (Continued)

RADIO FREQUENCY (RF) – Frequency intermediate between audio frequency and infrared frequencies. Used in magnetic resonance systems to excite nuclei to resonance. Typical frequency range for magnetic resonance systems is 5–130 Mhz.

RADIO FREQUENCY SHIELDING – Using material (e.g. copper, aluminium, or steel) to reduce interference from external radio frequencies. A radio frequency shielded room usually encloses the entire magnet room.

RESONANCE – A large amplitude vibration caused by a relative small periodic stimulus of the same or nearly the same period as the natural vibration period of the system. In magnetic resonance imaging, the radio frequency pulses are the periodic stimuli which are at the same vibration period as the hydrogen nuclei being imaged.

SECURITY ZONE – Area within the Magnet Room where the magnet is located. Signs are posted outside the Magnet Room warning personnel of the high magnetic field existing in the Magnet Room and the possibility of ferrous objects becoming dangerous projectiles within this zone.

SHIELD COOLER COLD HEAD – An external refrigeration device which maintains the shields inside the cryostat at a constant temperature.

SHIM COILS – Shim coils are used to provide auxiliary magnetic fields in order to compensate for inhomogeneities in the main magnetic field due to imperfections in the manufacturing of the magnet or affects of steel in the surrounding environment.

SHIMMING – Correction of inhomogeneity of the main magnetic field due to imperfections in the magnet or to the presence of external ferromagnetic objects.

SUPERCONDUCTING MAGNET – A magnet whose magnetic field originates from current flowing through a superconductor. Such a magnet is enclosed in a cryostat.

SUPERCONDUCTOR – A substance whose electrical resistance essentially disappears at temperatures near zero Kelvin. A commonly used superconductor in magnetic resonance imaging system magnets is niobium–titanium embedded in a copper matrix.

TESLA (T) – The internationally accepted unit of magnetic flux density. One tesla is equal to 10,000 gauss. One milli Tesla is equal to 10 gauss.

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a magnetic resonance system. "Preinstallation work" is done to prepare the customer's premises for the installation of the products sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as: the facility input power line to the power distribution unit, as well as emergency power lines and facility power lines to the the Magnet Room. The electrical contractor shall ring out and tag all wires at both ends. Color–coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non–electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Installation of RF shielding in Magnet Room and installation of equipment anchors.
- RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements.
- Site construction or renovation.
- Installation of structural reinforcements as required.
- Installation of selected magnetic shielding as required.
- Installation of water treatment equipment if necessary.
- Scheduling of riggers to move magnet (under GE Medical Systems direction) into its final location within the Magnet Room.

All work MUST be in compliance with national and local building and safety codes.

A structural steel analysis may be necessary during site evaluation for a magnetic resonance system. All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Panel and Power Distribution Unit or the power lines required in the Magnet Room, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, acoustic treatment to contain acoustic noise to the RF shielded room, or the RF shielded room illustrated in this document.

INTRODUCTION (Continued)

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third–party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE's field engineers, personnel of third–party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre–installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of pre–installation information.

Note

Direction 2223170, Signa MR/*i* & CV/*i* Pre–Installation contains the data and requirements for planning for a new Signa 1.0T & 1.5T MR/*i* & CV/*i* system.

Note

Direction 2304880, Signa 1.5T Twin*Speed* Upgrade Pre–Installation contains the data and requirements for planning for a Signa 1.5T Twin*Speed* system Upgrade.

Note

Direction 2142460, Signa Horizon Upgrade Pre–Installation contains the data and requirements for planning for a Signa Horizon system Upgrade.

Note

Direction 2223751, Signa MR/*i* & CV/*i* Upgrade Pre–Installation contains the data and requirements for planning for a Signa MR/*i* or CV/*i* system Upgrade with WideOpen Enclosure for a system with an LCC Magnet.